European Journal of Applied Physiology (2022) 122:1709–1722
https://doi.org/10.1007/s00421-022-04953-z

ORIGINAL ARTICLE

Effects of an external pneumatic compression device vs static compression garment on peripheral circulation and markers of sports performance and recovery

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Received: 25 September 2021 / Accepted: 8 April 2022 / Published online: 27 April 2022
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Abstract

Purpose To identify the effects of a single 30 min partial lower leg external pneumatic compression (EPC) treatment compared to a static compression (SC) garment or a no treatment control (CTL) on markers of recovery and performance following a muscle damaging protocol.

Methods Thirty healthy, active males (23 ± 3 years; 180.2 ± 9.0 cm; 81.6 ± 11.3 kg) performed 100 drop jumps from a 0.6 m box followed by a randomized, single 30 min treatment of either a partial lower leg EPC device worn below the knee and above the ankle (110 mmHg), SC garment (20–30 mmHg) covering the foot and calf just below the knee, or no treatment CTL, and then returned 24 and 48 h later. Participants were assessed for measures of muscle soreness, fatigue, hemodynamics, blood lactate, muscle thickness, circumferences, and performance assessments.

Results The drop jump protocol significantly increased muscle soreness ($p < 0.001$), fatigue ($p < 0.001$), blood flow ($p < 0.001$), hemoglobin ($p < 0.001$), and muscle oxygen saturation (SMO_2; $p < 0.001$). Countermovement jump and squat jump testing completed after treatment with either EPC, SC, or CTL revealed no differences for jump height between any condition. However, EPC treatment maintained consistent braking force and propulsive power measures across all timepoints for countermovement jump testing. EPC and SC treatment also led to better maintenance of squat jump performance for average relative propulsive force and power variables at 24 and 48 h compared to CTL.

Conclusions A single 30 min partial leg EPC treatment may lead to more consistent jump performance following a damaging bout of exercise.

Keywords Partial external pneumatic compression · Doppler ultrasound · Force plates · Drop jumps · Soreness and fatigue · Damage · Force · Power · Strength

Communicated by Nicolas Place.

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Abbreviations

EPC	External pneumatic compression
THb	Total Hemoglobin
HR	Heart rate
PPT	Pressure-to-pain threshold
RFD	Rate of force development
SC	Static compression
SMO_2	Muscle oxygen saturation

Introduction

Commercial lower limb external pneumatic compression (EPC) devices operate by inflating and deflating to specific pressures for a given period of time. The external pressure gradient of compression garments reduces the area for swelling, hemorrhage, and hematoma formation (Gurovich and Braith 2013; Martin et al. 2018). This mimics the rhythmic skeletal muscle squeezing action during exercise, thereby improving both blood and lymphatic circulation by shifting fluid from interstitial spaces into the vascular system (Rithalia 1991). These EPC devices are commonly utilized by athletic populations to aid in recovery from exercise in hopes of enhancing skeletal muscle adaptations (Haun et al. 2017). In this regard, EPC has been shown to improve blood flow (Martin et al. 2015a, c; Kephart et al. 2015), enhance oxygenation of skeletal muscle tissue (Haun et al. 2017), reduce blood lactate levels (Martin et al. 2015b), and promote greater range of motion (Haun et al. 2017). EPC has also shown similar effects to massage on circulation (Franklin et al. 2014) as well as promoting regeneration of skeletal muscle fibers (Shin and Sung 2015). To this point, different types and techniques of massage may have an effect on post-exercise performance and recovery from injuries; therefore, EPC may act as a mechanical modality to mimic similar responses (Weerapong et al. 2005).

Others have explored the impact of EPC following bouts of plyometric bounds/drop jumps, short sprints (10–30 m), intermittent repeated sprints, whole body resistance exercise protocols, Wingate cycling test, and up to 60 min steady state cycling (Martin et al. 2015b; Duffield et al. 2010; Cochrane et al. 2013; Keck et al. 2015; Marcello et al. 2019; Weltman and Regan 1983). Post-exercise interventions with EPC have shown that vertical jump performance remains unchanged compared to control conditions, while no significant differences have been reported between EPC and control conditions for several contraction types (isokinetic, concentric, and eccentric) or velocities (concentric and eccentric) in active males (Haun et al. 2017; Cochrane et al. 2013).

In addition, other compression options are commercially available, such as static compression garments, which are also commonly utilized in athletic populations. These devices such as static compression socks have been reported to improve calf muscle oxygenation and venous function during resting values (Bringard et al. 2006), without altering leg muscle blood flow velocity (Book et al. 2016). Furthermore, lower limb static compression tights have shown to reduce decrements in vertical jump and squat jump performance after 100 plyometric drop jumps (Jakeman et al. 2010). Similarly, Duffield et al. (Duffield et al. 2010) has shown reductions in levels of perceived muscle soreness following 24 h post-exercise, while wearing lower leg static compression garments during and for 24 h following high intensity sprint and plyometric exercise. Whereas static compression socks, which only cover the region below the knee have shown no effect in muscle soreness, calf circumference, or ankle dorsiflexion following a damaging eccentric calf exercise protocol (Heiss et al. 2018).

The most significant post-exercise EPC treatment effects are changes that occur in skeletal muscle blood flow and soreness (Martin et al. 2018). In addition, in a cohort of elite athletes (e.g., triathlon, wrestling, gymnastics, weightlifting) an increase in the pressure-to-pain threshold (PPT), measured via algometry, following an EPC treatment between two training sessions has been reported (Sands et al. 2015a). Measuring the PPT over multiple days, showed that PPT was significantly lower in the full leg EPC treatment group 3–6 days after completing a back squat exercise bout (10 sets of 5 repetitions at 80% 1RM) compared to a control condition (Haun et al. 2017).

Most if not all leg EPC devices cover the entire leg, but to our knowledge, there are currently no studies examining the effects of a partial leg (worn over the gastrocnemius between the knee and ankle) dynamic EPC device on recovery from exercise. Therefore, it is possible that partial leg EPC may have different recovery effects on performance, since the muscles in the calf (*Gastrocnemius* and *Soleus*) contribute to the action of propelling the body up and off the ground. Companies claim that external pneumatic compression devices aid recovery and improve performance; therefore, further research is required to investigate the effectiveness of EPC via parameters, such as decreased blood lactate, improved blood flow, and decreased post-exercise swelling and soreness. To our knowledge, no studies have examined the impact of partial leg EPC treatments on the adaptations to jump height, peak torque production, or muscular strength. Therefore, the purpose of this study was to determine the effects of a partial lower leg EPC recovery device compared to a static compression sock (SC) and no treatment control (CTL) during muscle recovery following a damaging exercise protocol in its ability to promote recovery and maintenance of physical performance. We hypothesized that soreness and performance would differ for the EPC group compared to SC and CTL.

Methods

Participants

Thirty active males (age 23 ± 3 years; height 180.2 ± 9.0 cm; body mass 81.62 ± 11.3 kg) were recruited from Lindenwood University and the surrounding area. Participants were included if they were: (a) exercising a minimum of 3 days per week; (b) 18–30 years of age; (c) BMI < 30 kg/m^2; (d) free of any known cardiovascular, respiratory, metabolic, or musculoskeletal ailments, and (e) were not taking any supplements or ergogenic aids. Prior to initiating the study, participants were informed of the experimental procedures, read, and signed informed consent forms, and filled out a medical history questionnaire. Notably, all consent and experimental procedures were approved through the Lindenwood University Institutional Review Board (IRB ID: 19–23, approval date: 07/17/2019), and conformed to the standards set by the latest revision of the Declaration of Helsinki.

Experimental design

The study was conducted using a randomized parallel group design, whereby participants completed a familiarization visit (visit 1), where an informed consent was voluntarily obtained, a DEXA scan was performed, and countermovement and squat jumps were practiced. Participants were then instructed to fast for 8 h and to refrain from exercise, alcohol, caffeine, and supplements for 72 h prior to visit 2. In addition, participants were instructed to keep a similar nutritional intake for the duration of the study and were given a commercially available 355 mL beverage containing 24 g CHO (Gatorade, Pepsico, Harrison, NY USA) to consume two hours prior to reporting for visit 2.

Upon arrival for visit 2, baseline measurements of the following pre exercise assessments were performed: hydration, body weight, resting heart rate (HR), resting blood pressure (BP), blood lactate, blood flow, muscle thickness, muscle circumferences, measures of fatigue and soreness, and performance measures. Upon completion of these measures, an exercise protocol consisting of 100 plyometric drop jumps ensued. 5 min following the muscle damage/exercise protocol, participants completed muscle fatigue and soreness measures, and blood lactate was measured. Participants were then randomized to their respective treatment condition using a simple randomization procedure via random.org. During the next 30-min participants received one of the three possible treatments described further below. At approximately 15 and 30 min after commencement of treatment, femoral artery blood flow and blood lactate concentration were assessed. Following the aforementioned assessments after 30 min of treatment measurements of thigh and calf thickness and circumference, participants rated their fatigue as well as their calf and thigh soreness using a visual analog scale, PPT of the *Vastus lateralis* and *gastrocnemius* via algometry, five maximal isokinetic plantar and dorsiflexion movements, three countermovement jumps and three squat jumps on force plates were completed. Throughout visit 2 skeletal muscle oxygen saturation was analyzed with a near infrared spectroscopy sensor at the midpoint of the right thigh.

Visits 3 and 4 occurred approximately 24 and 48 h after the completion of visit 2 and participants reported to the laboratory after following identical instructions as previous visits (fasting, abstention from caffeine, alcohol, nicotine, and exercise). The visits consisted of assessments of hydration, body weight, resting HR, resting BP, blood lactate, thickness of the *Vastus lateralis* and *gastrocnemius*, circumference of thigh and calf, participants rated their fatigue as well as their calf and thigh soreness using a visual analog scale, and PPT of the *Vastus lateralis* and *gastrocnemius* via algometry. Participants then completed maximal isokinetic ankle dorsi- and plantarflexion and three countermovement jumps and squat jumps at maximal effort.

Procedures

Urine specific gravity

Using a handheld refractometer and prior to each visit, participants provided a mid-stream urine sample for determination of their urine specific gravity and assessment of hydration status. Urine specific gravity values above 1.020 were deemed as a dehydrated sample. Participants were required to remain in the lab consuming water until their urine specific gravity sample was below 1.020 (Casa et al. 2000).

Body composition

At the beginning of each visit body mass was measured for every participant without shoes and in minimal clothing using a self-calibrating digital scale (Tanita, model # BWB-627A Class III, Tokyo, Japan) and at visit 1 height was measured with a standardized wall-mounted stadiometer (Tanita, model # HR-200, Tokyo, Japan). All participants underwent a whole-body scan using a Hologic QDR Discovery A DEXA scanner (Waltham, MA, USA) and body composition metrics were recorded for descriptive purposes and to ensure no between group differences, which there were none.

Heart rate and blood pressure measurement

HR, systolic (SBP), and diastolic (DBP) BP were measured during visits 2, 3, and 4 after participants rested in a supine position in a quiet room for 10 min prior to measurement. HR, SBP, and DBP measurements were taken at the brachial artery of the right arm using an automated oscillometric device (OMRON BP785, Omron Corporation, Kyoto, Japan).

Assessments of soreness and fatigue

PPT was assessed using a manual algometer (Force Ten, FDX 25: Wagner Instruments, Greenwich, CT, USA). The algometer pressure transducer was applied to the point at which the pressure became uncomfortable (audibly indicated by participants) which was termed PPT and the value was recorded in Newtons (N). All measures were taken in a seated position at the mid-thigh located at point between the iliac crest and patella of the right femur, and the *medial gastrocnemius* taken from the region of the largest circumference via visual inspection. The PPT was measured sequentially at the same site (mid-thigh and medial gastrocnemius) three times for triplicate measures with 10 s rest between measurements. The average of the triplicate measures at each site was calculated as the respective PPT. Notably, the digital display of the algometer indicating force of application was blinded to the participants. Measurements were performed prior to exercise, post-exercise, post-treatment, 24 and 48 h post-treatment.

Participants were asked to rate pain intensity by placing a mark on a 10 cm visual–analog scale. The visual–analog scale was horizontally positioned with a section referring to their current soreness level with the extremes labeled "no soreness" and "painful movement", their current level of fatigue with the extremes labeled "no fatigue and "exhausted", and their perceived difficulty of the exercise bout with the extremes labeled "no challenge" and "extremely challenging". Measurements were performed prior to exercise, post-exercise, post-treatment, 24 and 48 h post-treatment; however, the third question (difficulty of exercise bout) was only asked after completing the exercise bout.

Blood lactate

Blood samples were taken in laboratory conditions from the participants fingertip, where the fingertip site was cleaned, disinfected, and allowed to dry prior to puncture with a lancet. The first drop of blood was wiped away and the second drop of blood was placed on a strip and analyzed within a few seconds using a handheld portable analyzer Lactate Plus (Nova Biomedical, MA, USA). Samples were evaluated prior to exercise, post-exercise, 15 min post-treatment, 30 min post-treatment, 24 and 48 h post-treatment.

Muscle thickness

Participants were subjected to an ultrasound assessment to determine right leg mid-thigh muscle thickness of *Vastus lateralis* and right leg *medial gastrocnemius* muscle thickness with a 3 to 12 MHz multi-frequency linear phase array transducer (Logiq e BT12; General Electric, Fairfield, CT, USA). Mid-thigh measurements were taken from the midway point between the iliac crest and patella of the right femur, and *medial gastrocnemius* measures were taken from the region of the largest circumference via visual inspection. Participants were in a standing position and were instructed to shift a majority of the weight onto the left leg without the right leg leaving the ground. Reliability for muscle thickness during a test–retest at PRE on 33 participants produced an ICC of 0.994 (Haun et al. 2018). Measurements were performed prior to exercise, post-treatment, 24 and 48 h post-treatment.

Circumference measurements

Markings for the thigh were made between the midpoint of the right greater trochanter and the border of the right patella. Markings for the *gastrocnemius* were made at the largest medial midpoint protrusion of the right *gastrocnemius* muscle. Markings were made with a permanent marker and remained visible throughout the duration of the study for each participant with an occasional "touch-up" of the marks when fading occurred. Thigh and calf circumferences were measured on the right side of the body in a standing position with a flexible tape measure at the aforementioned markings prior to exercise, post-treatment, 24 and 48 h post-treatment.

Femoral artery blood flow measurement

Blood flow assessments were completed using a previously established protocol (Martin et al. 2017). Immediately following circumference measures, blood flow was assessed in a supine position on the right common femoral artery using an ultrasound (Logiq e BT12; General Electric, Fairfield, CT, USA) with a 3 to 12 MHz multi-frequency linear phase array transducer. The femoral artery was imaged longitudinally with the transducer placed 2–3 cm proximal to the bifurcation of femoral artery. Simultaneous measurement of artery diameter and blood velocity was performed using duplex mode imaging (B-mode and Doppler) and video was captured through a digital interface at 30 frames/s with real time analysis (FMD Studio, Pisa, Italy). Measurements were captured for one continuous minute with the transducer held manually in the same position. Vessel diameters were

determined frame-by-frame via automatic edge detection software (FMD Studio, Pisa, Italy) measuring the distance between the near and far wall of the intima. Blood velocity was determined via selection of a region of interest around the doppler waveform and a trace of the velocity–time integral was used to calculate mean velocity for each cardiac cycle. Blood flow from continuous diameter and mean blood velocity measurements during ultrasonography were calculated as [$\Pi \times (\text{diameter}/2)^2 \times$ time average mean velocity $\times$ 60]. Measurements were performed prior to exercise, 15 min post-treatment, and 30 min post-treatment.

Near-infrared spectroscopy measures (NIRS)

At visit 2, participants were fitted with the near infrared spectroscopy sensor (Humon; Dynometrics Inc., Cambridge, MA, USA) during resting conditions and prior to exercise testing. Briefly, the NIRS device was applied on the right leg at the mid-thigh over the rectus femoris following a previously published protocol (Farzam et al. 2018). Relative changes in muscle oxygen saturation (SMO_2%) and total hemoglobin (THb) were measured continuously throughout the entire duration of the second visit. Data was averaged for one min at the following timepoints: prior to exercise, post-exercise, 15-min post-treatment, and 30 min post-treatment.

Performance assessments

A Biodex System 3 (Biodex Corp., Shirley, NY, USA) isokinetic dynamometer was used for all testing. Bioware (Biodex Corp., Shirley, NY) computer software was used for data recording and reduction. Prior to isokinetic plantar- and dorsiflexion at 60°/sec testing, each participant completed a familiarization visit, where setup position was recorded and replicated in each visit, and then performed practice repetitions. All assessments were completed on four occasions (pre-treatment, immediate post-treatment, 24 h post-treatment, and 48 h post-treatment) consisting of two warm-up and five maximal effort repetitions. The highest peak torque of the five reps was recorded and used herein.

Countermovement and squat jump testing were performed on a Hawkins Dynamic dual force plate system (Portland, ME, USA) with a sampling frequency of 1000 Hz. Participants were given verbal instructions to keep hands on the hips, jump off and land with both feet on the force plates, and provide maximal effort. For both countermovement and squat jumps participants completed two to three warm-up jumps followed by a 1-min rest period, and then performed three jumps separated by a 30 s rest in between each jump. Countermovement and squat jump testing was performed at visit 1 for familiarity, prior to beginning the drop jump protocol, 30 min post-treatment, and 24 and 48 h post-treatment. The average of the three attempts were taken and the following variables were analyzed for countermovement and squat jumps: propulsive and braking phase, positive and propulsive impulse, braking and propulsive rate of force development (RFD), peak and peak relative braking and propulsive force, average power, average propulsive power, landing stiffness, and average landing force.

Exercise bout

Participants performed 10 sets of 10 maximal drop jumps from a 0.6 m high box with 1 min of passive recovery between sets and 10 s between jumps. Participants alternated the foot stepping off the box with each set. Participants were instructed to jump as high as possible and the last 3 s of rest were verbally announced. This protocol has been used successfully to induce exercise induced muscle damage in previous studies (Twist and Eston 2005).

External pneumatic compression treatment, static compression treatment, or control

An external pneumatic compression (EPC) device with a dynamic wavetec compression pattern (MAS Design Inc, New York, NY, USA) was used in a lying supine position for treatments for 30 min. The device consists of two wireless sleeves which are worn over the *gastrocnemius* between the knee and ankle while combining three massage techniques: pulsing compression, gradient, and distal release. Per the manufactures the highest compression setting, the device reaches 90 mm Hg on the top of the sleeve (slightly below the knee), 100 mm Hg in the middle of the sleeve, and 110 mm Hg on the bottom of the sleeve.

The static compression (SC) treatment condition consisted of participants lying in a supine position while wearing continuous external static compression running socks 2.0 (Shitsett, NC, USA) for 30 min. The socks were identified for right and left foot, starting at the toes, and ending below the knee. Based on the participant's calf circumferences, the appropriate size was selected for the participant. The socks are made of 85% polyamide (nylon) and 15% elastane and are indicated by the manufacturer to provide 20–30 mmHg graduated compression.

The control (CTL) condition received no treatment following the exercise bout to serve as a true control, where participants lay comfortable in a supine position for 30 min.

Statistics

Statistical analysis was performed using the statistical software JASP Team (2020) (Version 0.14.1), and data presented herein are reported as the mean ± standard deviation (SD) for all dependent variables. Prior to statistical analysis, all data were checked for normality and each respective statistical

test's assumptions along with baseline differences between conditions via a one-way ANOVA. If data were deemed not normally distributed by Shapiro–Wilk's testing, then data were logarithmically transformed and reassessed for normality. Furthermore, if pre-exercise differences were examined between conditions then to account for these differences' pre-exercise measures were used as a covariate to control for those differences. All dependent variables were examined using a mixed (condition by time) factorial ANOVA. If the assumption of heteroscedasticity for repeated measures were violated, a Greenhouse–Geisser correction factor was applied. Due to sample size, an interaction was only decomposed if significant and was found to have a large (> 0.14) partial Eta squared (η_p^2) effect size. This was accomplished by a one-way ANOVA across time within each condition or across conditions at each timepoint, and *post-hoc* testing was conducted with Holm correction factors applied. Statistical significance for all null hypothesis testing was set at $p \leq 0.05$. In addition, for significant between conditions *post-hoc* comparisons 95% confidence intervals (CI) of the mean difference between conditions, and magnitude of the effect was calculated using Cohen d effect size, where 0.2, 0.5, and 0.8 were considered small, moderate and large effects.

Results

Exercise difficulty

There were no differences ($p = 0.980$) between conditions for self-reported ratings of exercise difficulty (CTL: 4.6 ± 1.7, SC: 4.6 ± 2.1, EPC: 4.8 ± 2.3) measured immediately following the exercise bout. Thus, indicating that the exercise bout was equally challenging for all conditions.

Hemodynamics—blood flow, muscle O_2 saturation, and hemoglobin

There were statistically significant main effects of time for blood flow ($p < 0.001$; Fig. 1a), THb ($p < 0.001$; Fig. 1b), and SMO_2 ($p < 0.001$; Fig. 1c). However, there were no statistically significant main effects of condition or a condition by time interactions ($p < 0.05$) for these measures.

Blood lactate clearance

Data was logarithmically transformed. There were no statistically significant main effects for time or condition, or a condition by time interaction ($p < 0.05$) for blood lactate measures (Fig. 1d).

Thickness and circumference measures

There were no statistically significant main effects for time or condition, or a condition by time interaction ($p < 0.05$) for mid-thigh thickness, calf thickness, mid-thigh circumference, or calf circumference measures (see Fig. 2).

Soreness and fatigue measures

There were statistically significant main effects of time for visual analog scale soreness ($p < 0.01$; Fig. 3a) and visual analog scale fatigue ($p < 0.01$; Fig. 3b). However, there were no statistically significant effects of condition or condition by time interactions ($p < 0.05$) for these measures.

Pre-exercise differences for PPT at the mid-thigh were examined between conditions, and to account for these differences' pre-exercise measures were used as a covariate for this model. There was no statistically significant main effects of condition, time, or a condition by time interaction ($p < 0.05$) for PPT at the mid-thigh (Fig. 3c) or for PPT at the gastrocnemius (Fig. 3d).

Performance measures—isokinetic dynamometer

Pre-exercise differences for plantar flexion peak torque were examined between conditions, and to account for these differences' pre-exercise measures were used as a covariate for this model. There was no statistically significant main effects of condition, time, or a condition by time interaction ($p < 0.05$) for plantar flexion peak torque production (Fig. 4a) or dorsi flexion peak torque production (Fig. 4b).

Performance measures—countermovement jump

There was a statistically significant large condition by time interaction effect for average relative braking force ($p = 0.018$, $\eta_p^2 = 0.193$; Fig. 5b). Post hoc analysis revealed no statistically significant differences between conditions at any timepoint measured. However, within the CTL and SC conditions average relative braking force was higher at pre-exercise compared to 24 (CTL: $p < 0.001$, SC: $p = 0.008$) and 48 h (CTL: $p < 0.001$, SC: $p = 0.008$), where there was no change across time within the EPC condition ($p < 0.05$).

There were statistically significant main effects of time for average relative braking power ($p = 0.013$; Fig. 5c), braking rate of force development ($p < 0.001$; Fig. 5d), braking phase ($p = 0.001$; Fig. 5g), and average relative propulsive force ($p = 0.006$; Fig. 5e). However, there were no statistically significant main effects for condition or a condition by time interaction for these measures ($p < 0.05$).

There was a statistically significant large condition by time interaction effect for average relative propulsive power ($p = 0.030$, $\eta_p^2 = 0.176$; Fig. 5f). Post hoc analysis revealed

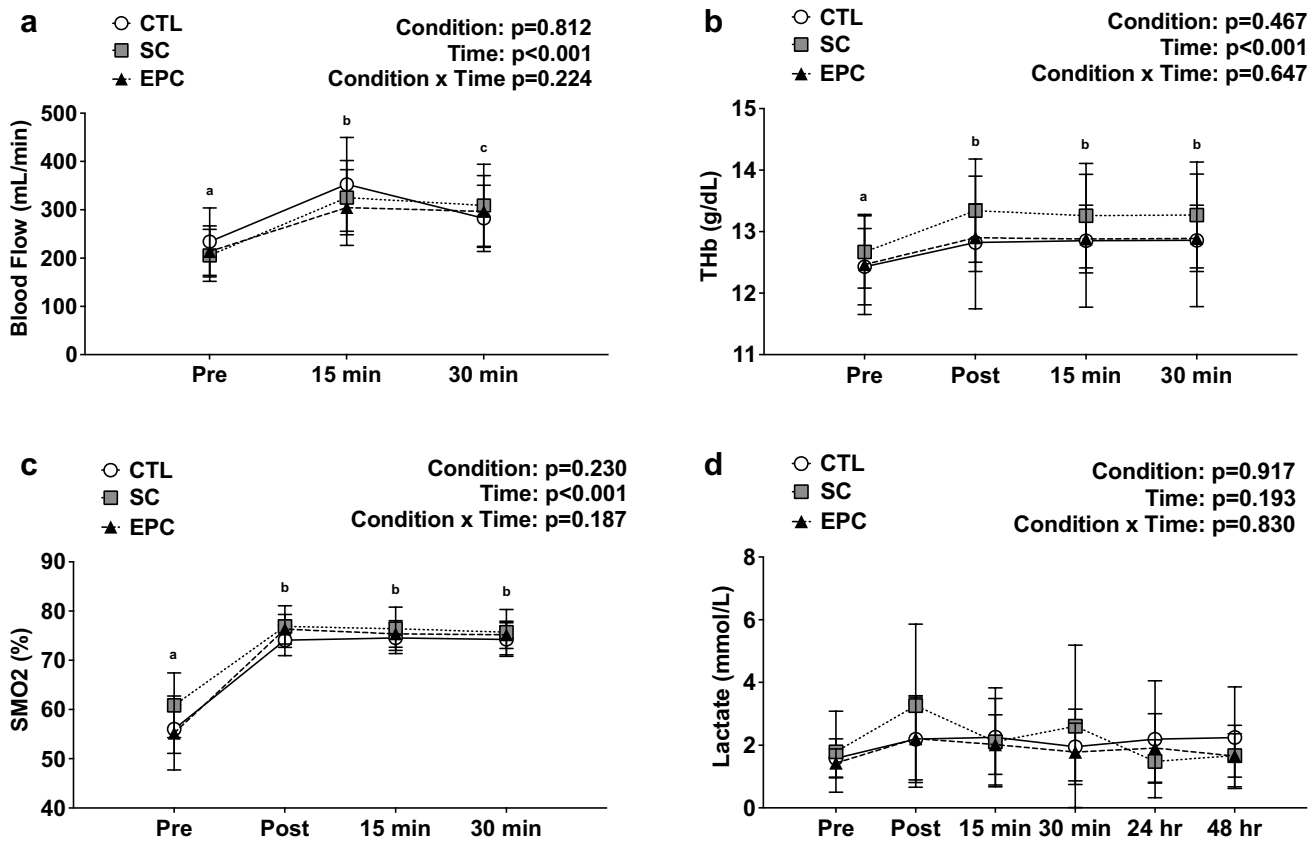


Fig. 1 Effects of either EPC, SC treatment, or CTL on hemodynamics and blood lactate. **a** Blood flow measures prior to exercise (pre), 15 min post-treatment (15 min) and 30 min post-treatment (30 min). **b** Total Hemoglobin (THb) measured prior to exercise (pre), 5 min after the drop jumps (post), 15 min post-treatment (15 min) and 30 min post-treatment (30 min). **c** Skeletal muscle oxygen percent (SMO_2%) was measured prior to exercise (pre), 5 min after the drop jumps (post), 15 min post-treatment (15 min) and 30 min post-treatment (30 min). **d** Blood lactate was measured prior to exercise (pre), 5 min after the drop jumps (post), 15 min post-treatment (15 min), 30 min post-treatment (30 min), 24 h post-drop jumps (24 h), and 48 h post-drop jumps (48 h). Different letters (e.g., a, b) represent differences between timepoints regardless of condition, where shared letters (e.g., a, a) represent no differences. Data are presented as mean ± standard deviation. EPC external pneumatic compression, SC static compression, CTL control

no statistically significant differences between conditions at any timepoint measured. However, within the CTL condition average relative braking power was higher at pre-exercise compared to 30 min post-treatment (CTL: $p = 0.007$), 24 (CTL: $p = 0.007$), and 48 h (CTL: $p = 0.007$), where no differences occurred over time within both the SC and EPC condition.

There were no statistically significant main effects for condition or a condition by time interaction for countermovement jump depth or jump height ($p < 0.05$).

Performance measures—squat jump

There was a statistically significant large condition by time interaction effect for squat jump average relative propulsive force ($p = 0.018$, $\eta_p^2 = 0.188$; Fig. 6b), and simple main effects occurred between conditions at 24 ($p = 0.009$) and 48 h ($p = 0.001$). Post hoc analysis revealed higher squat jump average relative propulsive force in both the EPC (159.36 ± 8.87 N·kg^{-1}; $p = 0.020$) and SC (159.78 ± 11.43 N·kg^{-1}; $p = 0.020$) conditions compared to the CTL condition (148.04 ± 5.34) at 24 h [EPC: 11.32 N·kg^{-1} (1.45 ± 21.20); SC: 11.74 N·kg^{-1} (1.87 ± 21.61 N·kg^{-1}); Fig. 6b] which resulted in large effects (EPC: $d = 1.55$; SC: $d = 1.32$). Post hoc analysis also revealed higher squat jump average relative propulsive force in both the EPC (160.89 ± 8.63 N·kg^{-1}; $p = 0.003$) and SC (161.05 ± 11.16 N·kg^{-1}; $p = 0.003$) conditions compared to the CTL condition (146.70 ± 5.88 N·kg^{-1}) at 48 h [EPC: 14.19 N·kg^{-1} (4.41 ± 23.97); SC: 14.35 N·kg^{-1} (4.57 ± 24.13); Fig. 6b] which resulted in large effects (EPC: $d = 1.92$; SC: $d = 1.61$). In addition, the CTL conditions measure of average relative propulsive force changed over time ($p < 0.001$), where pre- and post-measures were higher than 24 (Pre: $p = 0.004$; Post: $p = 0.001$) and 48 h

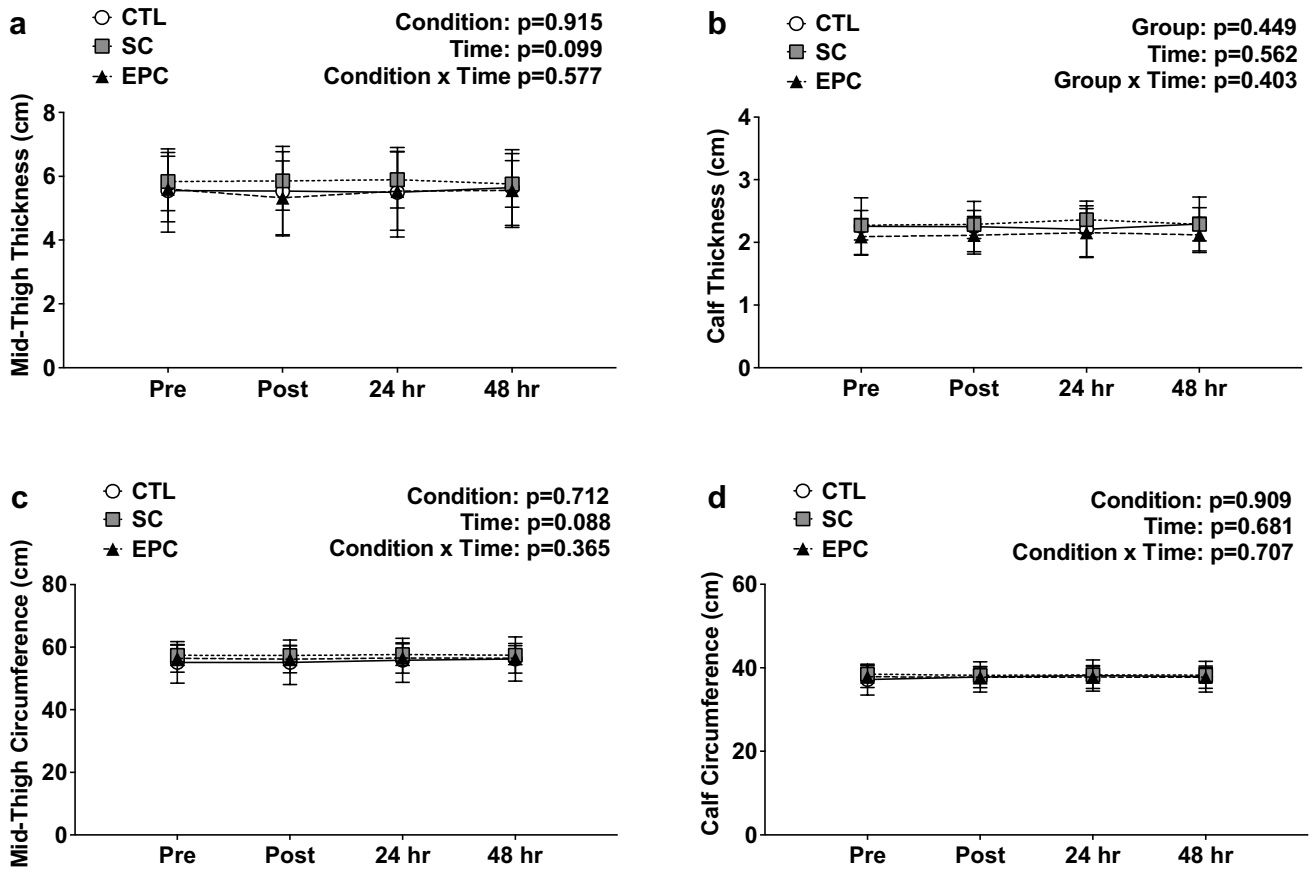


Fig. 2 Effects of either EPC, SC treatment, or CTL on skeletal muscle thickness or circumference. **a** Mid-thigh thickness, **b** calf thickness, **c** mid-thigh circumference, and **d** circumferences were measured prior to exercise (pre), 5 min after the drop jumps (post), 24 h post-drop jumps (24 h), and 48 h post-drop jumps (48 h). Different letters (e.g., a, b) represent differences between timepoints regardless of condition, where shared letters (e.g., a, a) represent no differences. Data are presented as mean ± standard deviation. EPC external pneumatic compression, SC static compression, CTL control

(Pre: $p = 0.001$; Post: $p = 0.011$). However, the EPC and SC condition did not change over time.

There was a statistically significant large condition by time interaction effect for squat jump average relative propulsive power ($p = 0.024$, $\eta_p^2 = 0.176$; Fig. 6c), and simple main effects occurred between conditions at 30 min post-treatment ($p = 0.050$), 24 ($p = 0.005$) and 48 h ($p < 0.001$). Post hoc analysis revealed higher squat jump average relative propulsive power in the EPC condition (18.99 ± 2.48 W·kg^{-1}) compared to the CTL condition (15.95 ± 2.50 W·kg^{-1}) at 30 min post-treatment [3.04 W·kg^{-1} (0.12 ± 5.96); Fig. 6c), which resulted in large effect ($d = 1.22$). In addition, post hoc analysis revealed higher squat jump average relative propulsive power in both the EPC (18.20 ± 2.88 W·kg^{-1}) and SC (17.24 ± 2.71 W·kg^{-1}) conditions compared to the CTL condition (14.11 ± 2.36 W·kg^{-1}) at 24 h [EPC: 4.08 W·kg^{-1} (1.14 ± 7.03); SC: 3.13 W·kg^{-1} (0.18 ± 6.08); Fig. 6c] which resulted in large effects (EPC: $d = 1.55$; SC: $d = 1.23$). Post hoc analysis revealed higher squat jump average relative propulsive power in both the EPC (18.76 ± 2.49 W·kg^{-1}) and SC (17.99 ± 2.30 W·kg^{-1}) conditions compared to the CTL condition (13.80 ± 2.47 W·kg^{-1}) at 48 h [EPC: 4.97 W·kg^{-1} (2.28 ± 7.65); SC: 4.19 W·kg^{-1} (1.51 ± 6.88); Fig. 6c] which resulted in large effects (EPC: $d = 2.00$; SC: $d = 1.76$). Furthermore, the CTL condition ($p < 0.001$) changed over time, where pre- and 30 min post-treatment measures were higher than 24 (pre: $p = 0.005$; Post: $p = 0.038$) and 48 h (pre: $p = 0.002$; Post: $p = 0.017$), but the EPC and SC condition did not change over time.

There were no statistically significant main effects for condition, time, or a condition by time interaction for squat jump propulsive RFD ($p < 0.05$).

There was a statistically significant main effect of time for squat jump height ($p = 0.009$; Fig. 6e). However, there were no statistically significant main effects for condition or a condition by time interaction for squat jump height ($p < 0.05$).

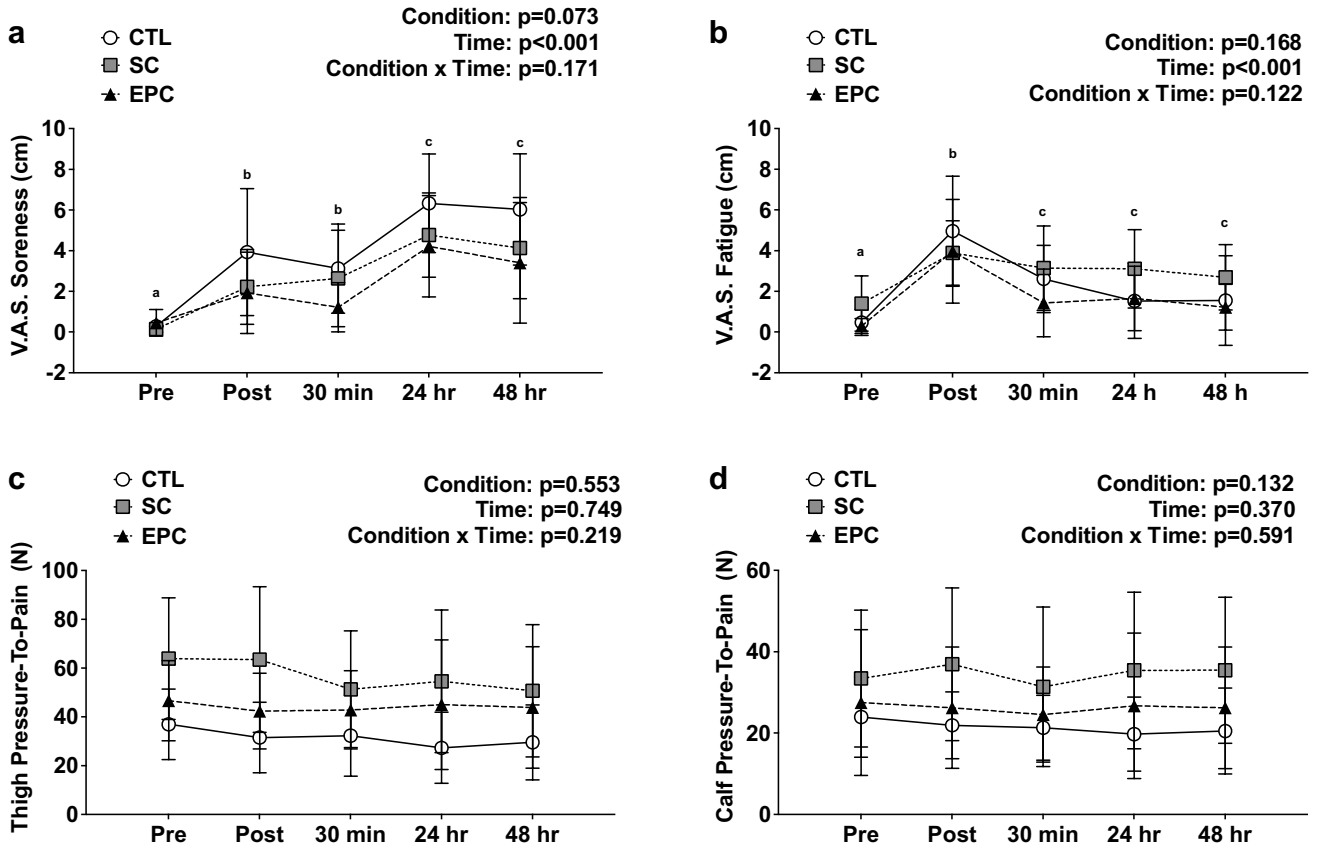


Fig. 3 Effects of either EPC, SC treatment, or CTL on perceived soreness, fatigue, and pressure-to-pain threshold. **a** VAS soreness, **b** VAS fatigue, **c** Thigh pressure-to-pain threshold, and **d** Calf pressure-to-pain threshold were measured prior to exercise (pre), 5 min after the drop jumps (post), 30 min post-treatment (30 min), 24 h post-drop jumps (24 h), and 48 h post-drop jumps (48 h). Different letters (e.g., a, b) represent differences between timepoints regardless of condition, where shared letters (e.g., a, a) represent no differences. Data are presented as mean ± standard deviation. EPC external pneumatic compression, SC static compression, CTL control, VAS visual analog scale

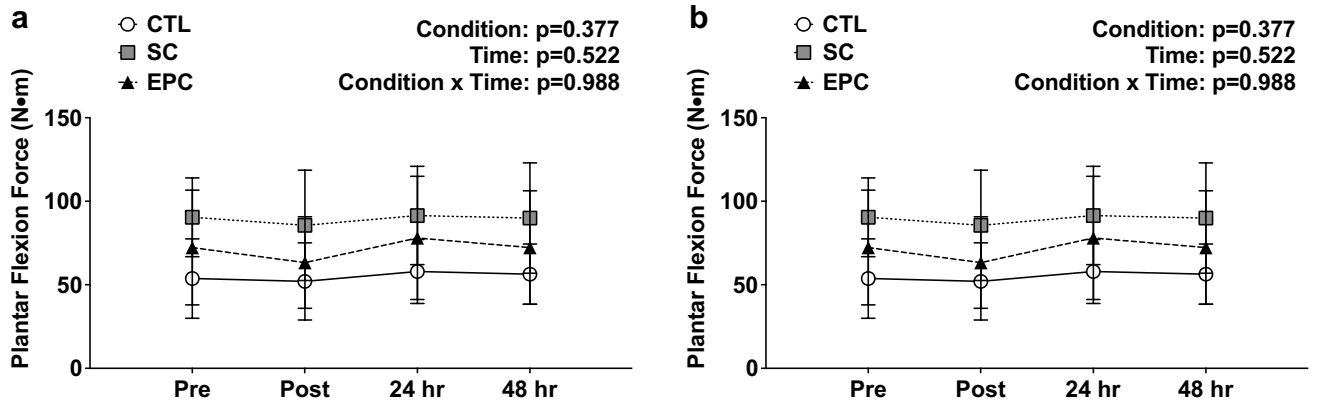
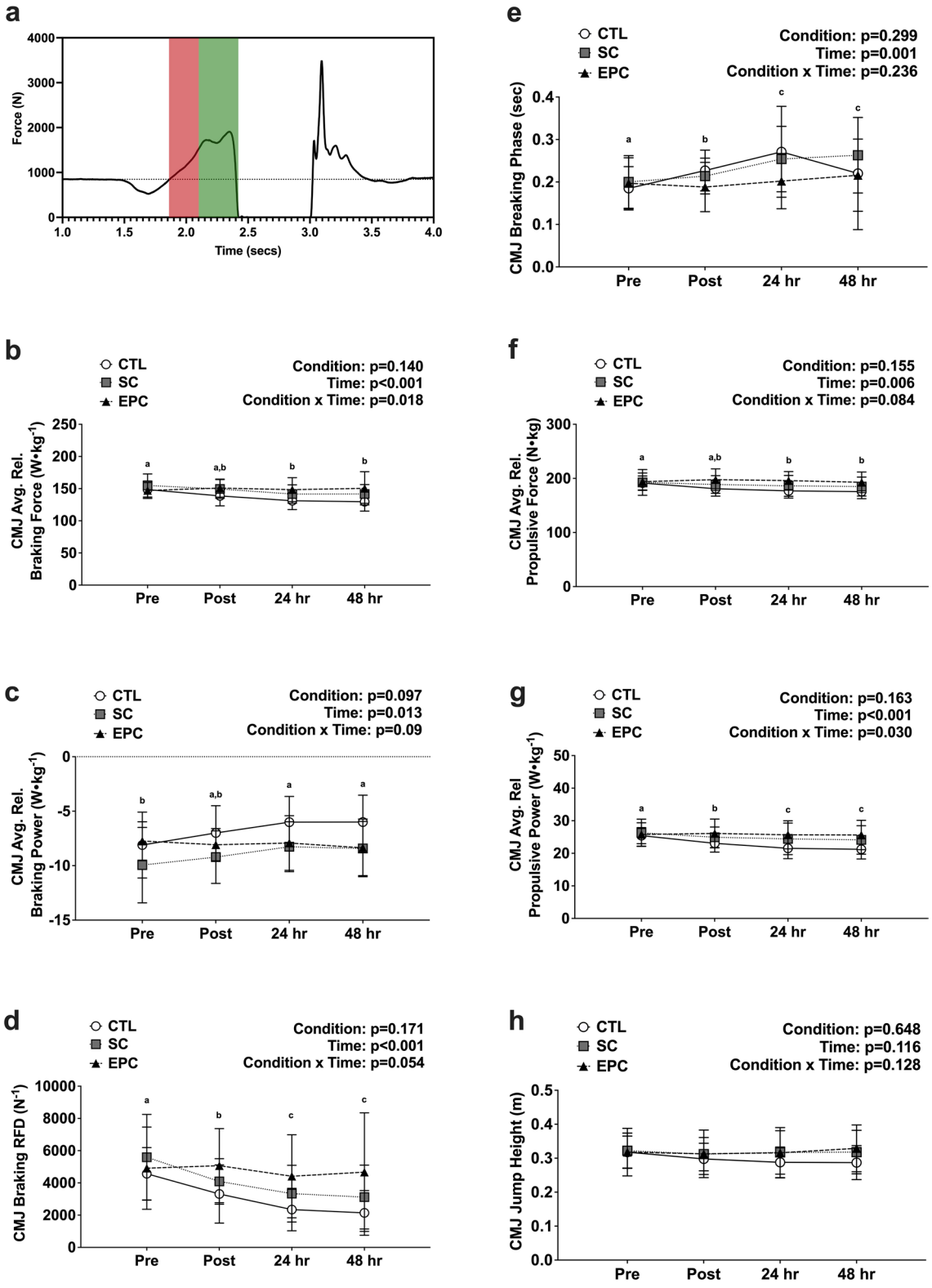


Fig. 4 Effects of either EPC, SC treatment, or CTL on isokinetic force production. **a** Plantar flexion force and **b** dorsi flexion force were measured prior to exercise (pre), after the completion of treatment (post), 24 h post-drop jumps (24 h), and 48 h post-drop jumps (48 h). Different letters (e.g., a, b) represent differences between timepoints regardless of condition, where shared letters (e.g., a, a) represent no differences. Data are presented as mean ± standard deviation. EPC external pneumatic compression, SC static compression, CTL control


a
Force (N)
Time (secs)
b
CTL
SC
EPC
Condition: p=0.140
Time: p<0.001
Condition x Time: p=0.018
CMJ Avg. Rel. Braking Force (W•kg-1)
Pre
Post
24 hr
48 hr
c
Condition: p=0.097
Time: p=0.013
Condition x Time: p=0.09
CMJ Avg. Rel. Braking Power (W•kg-1)
d
Condition: p=0.171
Time: p<0.001
Condition x Time: p=0.054
CMJ Braking RFD (N-1)
e
Condition: p=0.299
Time: p=0.001
Condition x Time: p=0.236
CMJ Breaking Phase (sec)
f
Condition: p=0.155
Time: p=0.006
Condition x Time: p=0.084
CMJ Avg. Rel. Propulsive Force (N•kg)
g
Condition: p=0.163
Time: p<0.001
Condition x Time: p=0.030
CMJ Avg. Rel Propulsive Power (W•kg-1)
h
Condition: p=0.648
Time: p=0.116
Condition x Time: p=0.128
CMJ Jump Height (m)

◂**Fig. 5** Effects of either EPC, SC treatment, or CTL on countermovement jump variables. **a** Force tracing, **b** average relative braking force, **c** average relative braking power, **d** braking RFD, **e** braking phase, **f** average relative propulsive force, **g** average relative propulsive power, and **h** jump height were measured prior to exercise (pre), after the completion of treatment (post), 24 h post-drop jumps (24 h), and 48 h post-drop jumps (48 h). The force tracing red color represents the braking phase, and the green represents the propulsive phase. Different letters (e.g., a, b) represent differences between timepoints regardless of condition, where shared letters (e.g., a, a) represent no differences. Data are presented as mean ± standard deviation. EPC external pneumatic compression, SC static compression, CTL control, CMJ countermovement jump, RFD rate of force development

Discussion

The aim of this study was to compare the effects of 30 min of partial leg EPC treatment, SC treatment, or a no treatment CTL following a muscle damaging exercise protocol while examining the ability to promote recovery and physical performance. Our findings suggest that partial leg EPC treatment may lead to more consistent performance when performing a jumping motion following a damaging exercise protocol.

Treatment effects on perceived soreness

Our findings of increased perceived soreness following a muscle damaging protocol are similar to prior research which report peak soreness 24–48 h following a damaging exercise bout (Jakeman et al. 2010; Vaile et al. 2008; Nosaka and Kuramata 1991; Kanik et al. 2019). However, we observed no differences between treatment conditions for perceived soreness, though numerically the EPC condition had lower perceived soreness following treatment. These findings are similar to previous research that have reported no differences with one hour of full leg EPC treatment (Draper et al. 2020) or 60 h of static compression sock treatment (Heiss et al. 2018) following muscle damaging exercise. Nonetheless, our results differed in comparison to acute 15 min full leg EPC treatment following aggressive training which was reported to lower muscle soreness (Sands et al. 2015b). Our findings also differ from Haun et al. who reported reductions in thigh PPT measures with one hour of full leg EPC treatment every day for 5 days after completing multiple resistance exercise bouts (Haun et al. 2017). Granted we only used a single 30 min post-exercise treatment herein, which potentially more frequent treatment may be needed to elicit reductions in muscle soreness. In addition, differences in soreness could have occurred after are last testing bout at 48 h as compression stockings have been

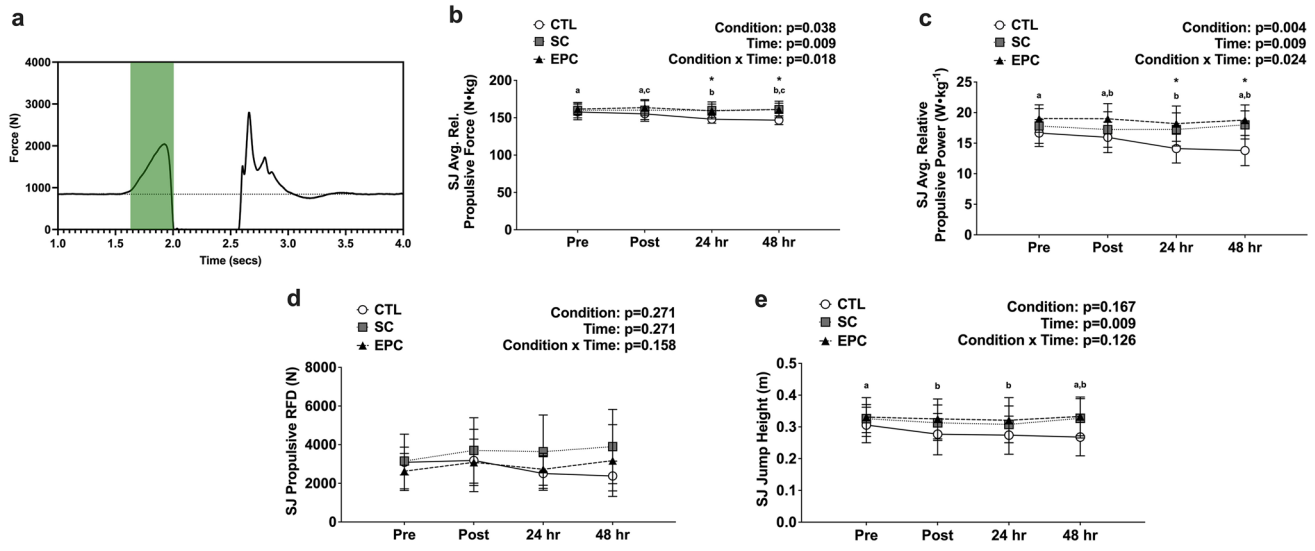


Fig. 6 Effects of either EPC, SC treatment, or CTL on squat jump variables. **a** Force tracing, **b** average relative propulsive force, **c** average relative propulsive power, **d** propulsive RFD, and **e** jump height were measured prior to exercise (pre), after the completion of treatment (post), 24 h post-drop jumps (24 h), and 48 h post-drop jumps (48 h). The force tracing green color represents the propulsive phase. Different letters (e.g., a, b) represent differences between timepoints regardless of condition, where shared letters (e.g., a, a) represent no differences. Asterisk represents differences between conditions, where both the EPC and SC were greater than CTL. Data are presented as mean ± standard deviation. EPC external pneumatic compression, SC static compression, CTL control, CMJ countermovement jump, RFD rate of force development

shown to reduce muscle soreness following 72 h of muscle damaging exercise (Perrey et al. 2008). The suggested mechanism to explain the causation of muscle soreness is muscle swelling elicited by muscle damaging exercise (French et al. 2008; Howatson and Milak 2009). However, we examined no changes in muscle swelling following the muscle damaging protocol. This differs from prior research using a similar protocol herein which did report findings of muscle swelling (Woolley et al. 2014).

Effects of EPC treatment on performance measures

Our results showed no differences between conditions for jump height during the 48 h recovery period. This agrees with findings by Cochrane et al. (Cochrane et al. 2013) who reported that neither intermittent pneumatic compression or SC changed single leg vertical jump performance. However, our findings differ from prior research (Jakeman et al. 2010) reported that full leg static compression treatment for 12 h was shown to improve vertical jump performance compared to their control group albeit the treatment time was drastically longer than the 30 min used within this study.

Notwithstanding, we observed changes both within the SC and CTL conditions for the braking portions of the countermovement jump, whereas the EPC treatment did not change over time. This could have been potentially better maintained with a partial leg EPC treatment and given that the drop jump places emphasis on the eccentric phase, this might be one explanation as to why the SC treatment and CTL conditions changed over time. Research has shown to minimize the braking phase, participants will have had to have a larger braking force, shorter time of impulse, and greater RFD throughout the braking phase (McMahon et al. 2017). We observed within the EPC condition consistent unimpaired performance within the braking portion of the countermovement jump. This may suggest that individuals within the EPC condition were able to perform the movement more efficiently throughout the study. In addition, large braking force requires a greater propulsion force to minimize the transition time between braking and propulsion phase leading to a faster reacceleration of the body mass (McMahon et al. 2017). To support this hypothesis, we observed similar favorable changes during the propulsive phase for EPC and SC conditions during the squat jump, which removes the eccentric braking portion of the countermovement jump. The removal of the braking phase when performing the squat jump would reduce the perception of soreness that occurred from the exercise protocol. In addition, the practical implications of these findings are limited to the muscle damaging protocol used herein and more research is needed to confirm these observations.

Effects of EPC treatment on hemodynamics

During the exercise bout blood flow increased and stayed elevated throughout the recovery period but did not differ between treatments. In this regard, full leg EPC treatment has been shown to enhance skeletal muscle blood flow (Martin et al. 2015a, c; Kephart et al. 2015), which differ from our current findings. This could potentially be due to full leg EPC treatment compressing larger volumes of tissue in the leg, and the treatment occurring closer to the measurement site. Whereas partial leg EPC treatment compresses less tissue and compression is occurring further away from the measurement site. In addition, the 30 min treatment period could possibly not be an adequate length of time using partial-leg EPC as other studies using full leg EPC have demonstrated benefits using 60 min or even multiple treatment sessions of EPC (Haun et al. 2017; Winke and Williamson 2018; Keck et al. 2015; Sands et al. 2015a). In accordance with our findings, previous investigations using NIRS devices have found that SMO_2 increased throughout exercise and remained elevated throughout recovery (Denis et al. 2011; Farzam et al. 2018). However, when examining the effects of full leg EPC treatment during normal resting conditions SMO_2 has been shown to increase after 60 min of treatment (Lassalle et al. 2019).

Limitations

In review, our observations are limited to the participants in this study, which were active and consistent exercisers, but these findings may differ in untrained or elite populations. Furthermore, our PPT measures were minimized to a specific muscle, whereas the self-reported perception of soreness and fatigue are relative to the overall sensation in the body, not localized on specific parts of the body which could have impacted our findings. Therefore, using a soreness report of the entire body or individual perceived soreness and fatigue scales for different muscles should be used in the future. Finally, the lack of increase in post-exercise blood lactate concentrations reported herein led to the inability to discern any treatment effect(s) and should be noted for future research investigations.

Conclusion

In conclusion, this is one of the first studies to our knowledge to compare the effects of 30 min of a partial leg EPC vs. SC treatment and a CTL. Our data suggest that a single 30 min partial leg EPC treatment may lead to more consistent jump performance following a damaging bout of exercise. Given that this is the first study examining the effects of partial leg EPC treatment we implore that more research is needed to

confirm these findings and to examine the effects of more strenuous, repeated exercise bouts, and longer treatments in different populations to determine the overall effects on recovery and performance.

Author contributions Conceptualization, J.S.M. and P.W.M.; methodology, J.C.B., S.R., K.L.S., C.M.K., J.S.M., and P.W.M.; data Collection, J.C.B., J.M.M, K.M.R., R.A.S., and P.W.M.; writing—original draft preparation, J.C.B. and P.W.M.; writing—review and editing, J.C.B., J.M.M., K.M.R., R.A.S., S.R., K.L.S., C.M.K., J.S.M., and P.W.M.

Funding Funding was provided to P.W.M by MAS Holdings (Colombo, Sri Lanka).

Declarations

Conflict of interest None of the authors have conflicts of interest to declare regarding the publication of these data.

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